

Seat No: _____

Enrollment No: _____

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Winter2023 - 24Examination

Semester: 3**Subject Code: 303105210****Subject Name: Computer Organization and Microprocessor (COM)****Date: 24/01/2024****Time: 10:30 am to 1:00 pm****Total Marks: 60****Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (Fill in the blanks, one word answer, MCQ-not more than Five in case of MCQ) (All are compulsory) (Each of one mark) (15)

1. ALE Stands for _____.
2. 1 nibble=_____bits.
3. SID means_____.
4. _____instruction is used to load 16-bit value in register pair.
5. Data bus is always _____.(bidirectional, unidirectional)
6. What stand for RISC and CISC?
7. What is T-state?
8. S0=1 and S1=1, then read operation performed by microprocessor –State True or False.
9. Explain function of Program counter.
10. 8085 is called 8 bit microprocessor- State True or False.
11. What is the size of stack in 8085 microprocessor?
a) 8 bit b) 10 bit c) 16 bit d) 12 bit
12. For 8085 microprocessor, if register B has 65 H and accumulator has 97 H, What will be the content of accumulator after executing instruction SUB B?
a) 65 H b) 97 H c) 32 H d) 28 H
13. The main purpose of Accumulator in 8085 is_____.
a) Temporary data storage b) Selection of peripherals
c) Storing instructions d) Used as primary pointer
14. Which of the following interrupts has highest priority in microprocessor 8085?
a) RST 7.5 b) RST 4.5 c) INTR d) TRAP
15. ASCII code stands for_____.
a) American Standard Code for Information Interchange
b) American Stable Code for International Interchange
c) American Standard Copy for Information Interchange
d) American Standard Case for Institutional Interchange

Q.2 Answer the following questions. (Attempt any three) (15)

- A) Give difference between microprocessor and microcontroller.
- B) Explain Vectored, Non vectored, Maskable and Non Maskable interrupts .Give examples also.
- C) Explain given Instructions: 1) MVI 2) CMP 3) XRA 4) SUB 5) STA
- D) Explain direct addressing mode and Indirect addressing mode of 8085with examples.

Q.3 A) Draw and explain Programming model of 8085. (07)

Q.3 B) Give classification of memory in 8085. (08)

OR

Q.3 B) Explain circular shift and arithmetic shift micro operations with examples. (08)

Q.4 A) Draw timing diagram of instruction MOV B, C. (07)
OR
Q.4 A) Draw pin diagram of 8085 microprocessor and explain READY and RESET IN pin. (07)
Q.4 B) Draw a common bus system using multiplexers. (08)